

BIOMECHANICS (EPHE 341)

Angular Kinetics 3

Newton's three laws, rewritten for things that turn — and the one quantity a person can change at will.

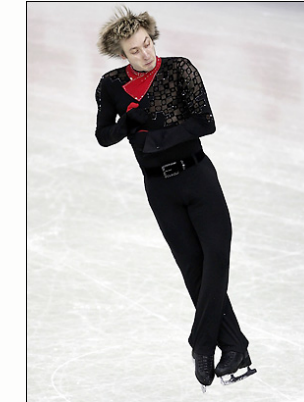
Angular Kinetics 3

The same three laws, said again for rotation

Every law from linear kinetics has an angular twin. The sentences are the same sentences with the nouns swapped.

	Linear	Angular
what resists	mass m	moment of inertia I
what acts	force F	torque M
1st law	constant v unless a force acts	constant ω unless a torque acts
2nd law	$F = ma$	$M = I\alpha$
momentum	$p = mv$	$L = I\omega$
impulse	$Ft = \Delta p$	$Mt = \Delta L$

One row is not a swap. **Mass is a number; moment of inertia is a number per axis**, and it changes when you move. That one difference is what the rest of this lecture is about.



Newton's first law, for turning

Linear. A body maintains a constant state of rest or constant velocity unless acted upon by an external *force*.

Angular. A body maintains a constant state of angular motion unless acted upon by an external *torque*.

Both halves matter, and the second half is the one people forget. “At rest” is a constant state of angular motion. So is spinning at eleven revolutions a second. Neither changes by itself.

In linear kinetics, *inertia* is quantified by mass — the resistance to a change in the state of motion.

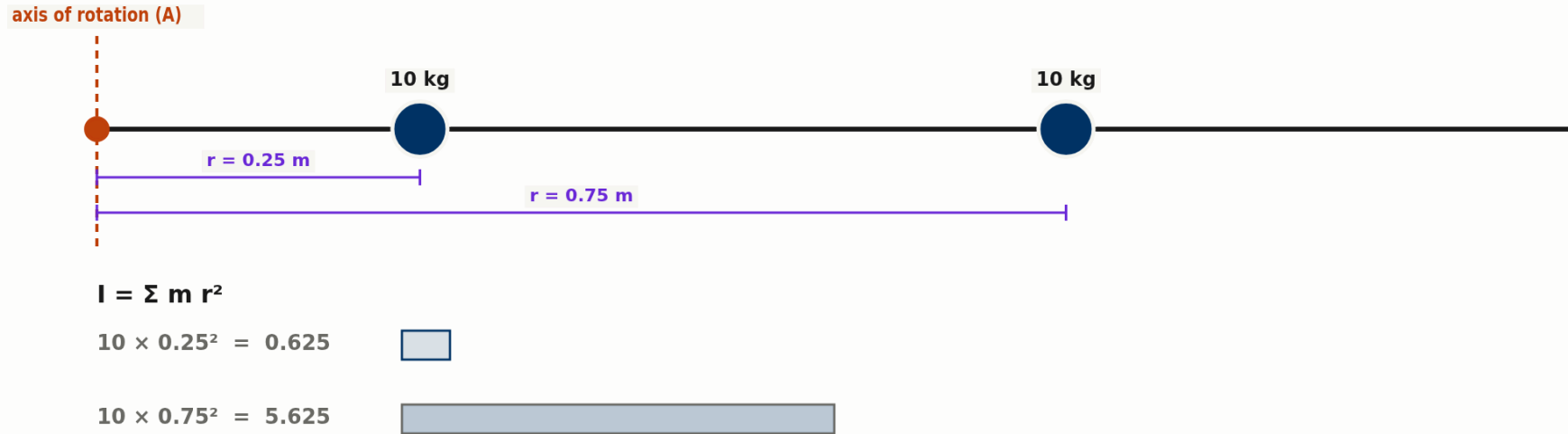
Angular inertia is the property of an object to resist changes in its *angular* motion. It depends on two things:

- the mass of the object
- **the distribution of that mass relative to the axis of rotation**

The second one is new. Nothing in linear kinetics cared *where* the mass was. Everything in angular kinetics does.

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Moment of inertia: $I = \Sigma m r^2$



$I = 6.250 \text{ kg} \cdot \text{m}^2$ his slide 7
Drag either mass. The far mass is carrying 90% of the whole moment of inertia — distance is squared in the sum, mass is not, which is the whole of his slide 6.

His example

Move one out

Double the mass instead

About the end (A)

About the centre of gravity

Every mass in an object resists being turned, and how hard it resists depends on **how far out it is, squared**. Drag the masses and watch the two terms in the sum change: move one out and its contribution grows four times as fast as its distance does. Then switch the axis to the centre of gravity and watch the total **fall** — which is the next slide.

What the squaring means

$$I = \sum m r^2$$
$$= m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + \dots$$

Mass appears once. Distance appears twice. So:

- double a mass and its contribution **doubles**
- double its distance and its contribution **quadruples**

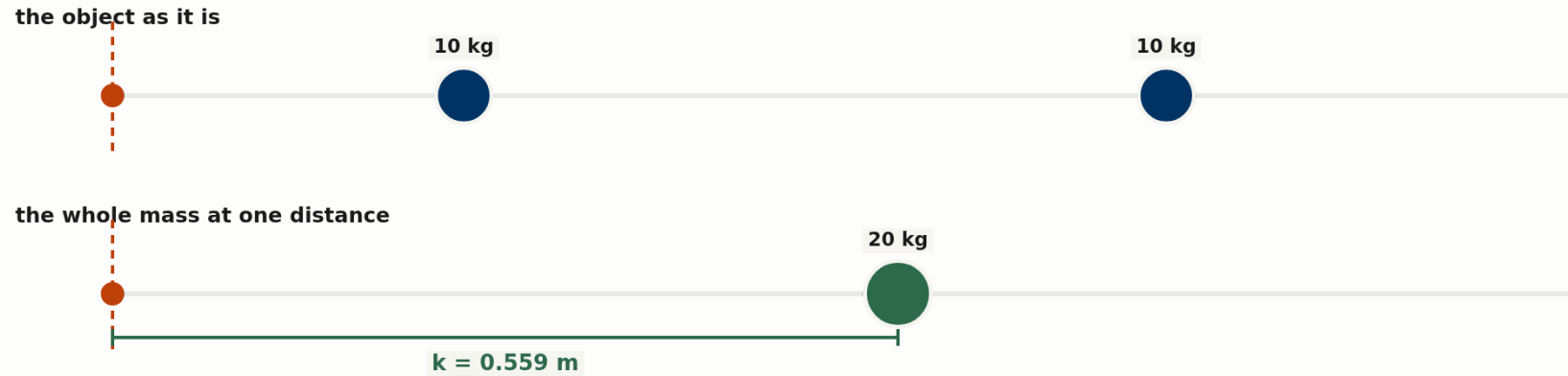
Which is why his slide says, in one line, that *the distribution of mass has a greater influence on angular inertia than mass*.

The units follow from the definition and are worth saying out loud once: kilograms times metres squared, **kg·m²**. Not kilograms. Not newtons. A moment of inertia is not a force and not a mass.

A useful habit: whenever you meet a new rotational quantity, check its units against the linear one it replaces.

$m \rightarrow \text{kg}$ becomes $I \rightarrow \text{kg} \cdot \text{m}^2$; $p \rightarrow \text{kg} \cdot \text{m/s}$ becomes $L \rightarrow \text{kg} \cdot \text{m}^2/\text{s}$; $Ft \rightarrow \text{N} \cdot \text{s}$ becomes $Mt \rightarrow \text{N} \cdot \text{m} \cdot \text{s}$. The extra metre is the moment arm, every time.

The radius of gyration



$$I = m k^2 \rightarrow k = \sqrt{I / m}$$

$k = \sqrt{(6.250 / 20)} = 0.559 \text{ m}$
 The radius of gyration is a distance, not a mass and not an inertia: it is where the whole 20 kg would have to sit to be exactly as hard to turn as the two masses actually are.
 Both have have $I = 6.250 \text{ kg} \cdot \text{m}^2$ about that axis.

About the end (A)

About the centre of gravity

Two 10 kg masses, 0.25 m and 0.75 m from the end. About the end they give $6.25 \text{ kg} \cdot \text{m}^2$, and the whole 20 kg would have to sit **0.559 m** out to match that. About the centre of gravity the same object gives $1.25 \text{ kg} \cdot \text{m}^2$ and **$k = 0.250 \text{ m}$** — the same object, a quarter of the resistance, because the axis moved.

Two ways to write the same number

$$I = \sum m r^2$$

every particle, counted separately

$$I = m k^2$$

all the mass at one distance k

Both give the same I . The second is how anthropometric tables report segment data: a mass fraction and a **radius of gyration expressed as a fraction of segment length**, so one table serves every body.

Winter's table gives three radii of gyration for each segment — about its own centre of gravity, about its proximal end, and about its distal end — because which one you need depends on where the axis is.

segment	mass	k_{CofG}	k_{prox}	k_{dist}
Thigh	0.100 M	0.323 L	0.540 L	0.653 L
Leg	0.0465 M	0.302 L	0.528 L	0.643 L
Foot	0.0145 M	0.475 L	0.690 L	0.690 L

The three are not independent: $k_{\text{prox}}^2 = k_{\text{CofG}}^2 + d^2$, which is the next slide.

Why the centre of gravity

Of all the axes an object could turn about, one of them is special — and it is special for a reason you can compute.

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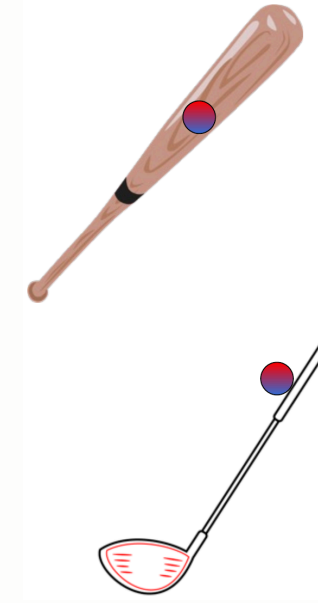
Anything thrown turns about its own centre of gravity

We established last lecture that an object in the air and free to rotate turns about its centre of gravity. Now we can add the second half:

- its resistance to turning is its **moment of inertia**
- which depends on its **mass** and on the **distribution** of that mass

A bat and a racquet are nearly the same length and nothing like the same object to swing, because one has most of its mass near the far end and the other has a strung hoop and a light shaft.

Thrown, both of them tumble about the marked point and nothing else — whatever shape they are, and however you throw them.



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Example problem — about the centre of gravity

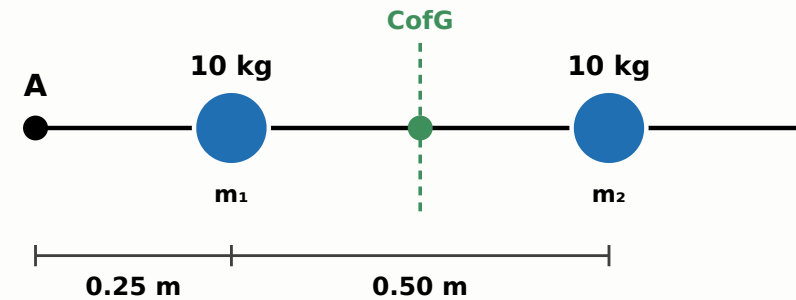
Same object: a 10 kg mass 0.25 m from point A, and a 10 kg mass 0.75 m from A. This time, find I about its **centre of gravity**.

First, locate the centre of gravity — last lecture's sum:

$$\begin{aligned}\text{CofG} &= \Sigma(m \cdot r) / \Sigma m \\ &= [(10)(0.25) + (10)(0.75)] / (10 + 10) \\ &= 10.0 / 20 = \mathbf{0.50 \text{ m}} \text{ from A}\end{aligned}$$

Then the moment of inertia about that point, with both masses now 0.25 m away:

$$\begin{aligned}I &= (10)(0.25)^2 + (10)(0.25)^2 \\ &= 0.625 + 0.625 = \mathbf{1.25 \text{ kg}\cdot\text{m}^2}\end{aligned}$$



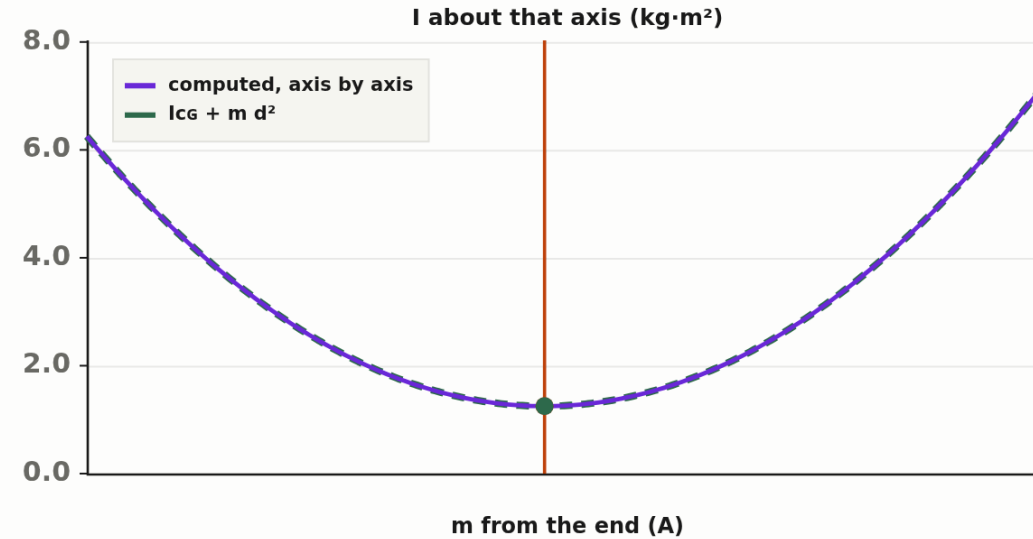
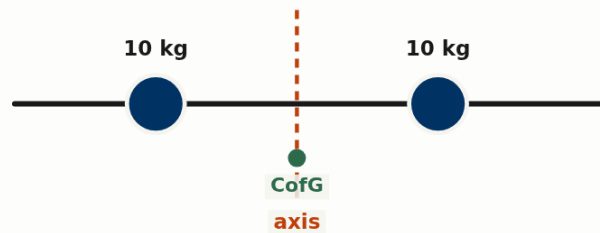
And the radius of gyration about the centre of gravity:

$$k = \sqrt{I/m} = \sqrt{1.25 / 20} = \mathbf{0.25 \text{ m}}$$

About A it was **6.25**. About the centre of gravity it is **1.25** — a fifth of the resistance, from the same object. Nothing moved but the axis.

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So why does a free object turn about its centre of gravity?



axis at 0.50 $\rightarrow I = 1.25 \text{ kg}\cdot\text{m}^2$ · smallest at the CofG, 0.50, where $I = 1.25$

his two masses · 20 kg · the two lines are a computation and a prediction, and they lie on top of each other because the parallel-axis theorem is exact

Where the
axis is



His two masses

A measured body

► Sweep the axis

Slide the axis anywhere you like and the moment of inertia traces a **parabola with its lowest point at the centre of gravity**. That is not a property of these two masses — switch to the measured body and the same thing happens, for the same reason: moving the axis a distance d away from the centre of gravity adds $m d^2$ and nothing else.

The parallel-axis theorem

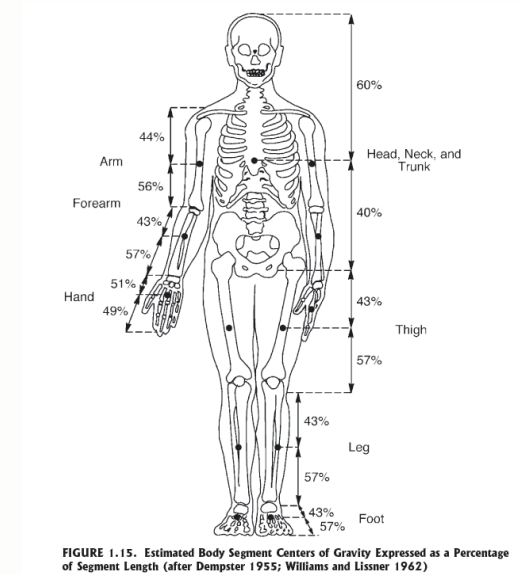
$$I_{\text{axis}} = I_{\text{CofG}} + m d^2$$

where d is the perpendicular distance from the centre of gravity to the axis.

It does three jobs at once:

- it explains the parabola on the last slide
- it says the minimum is always at $d = 0$
- it is how a table of segment data covers every axis you might need

Check it on the worked example: $1.25 + (20)(0.50)^2 = 1.25 + 5.00 = \mathbf{6.25}$, which is I about A.



Winter's table gives k about the segment's own centre of gravity. Everything else — about the hip, about the whole body — is that number plus $m d^2$.

A body that can change its own inertia

Mass you are stuck with. Where you hold it, you are not.

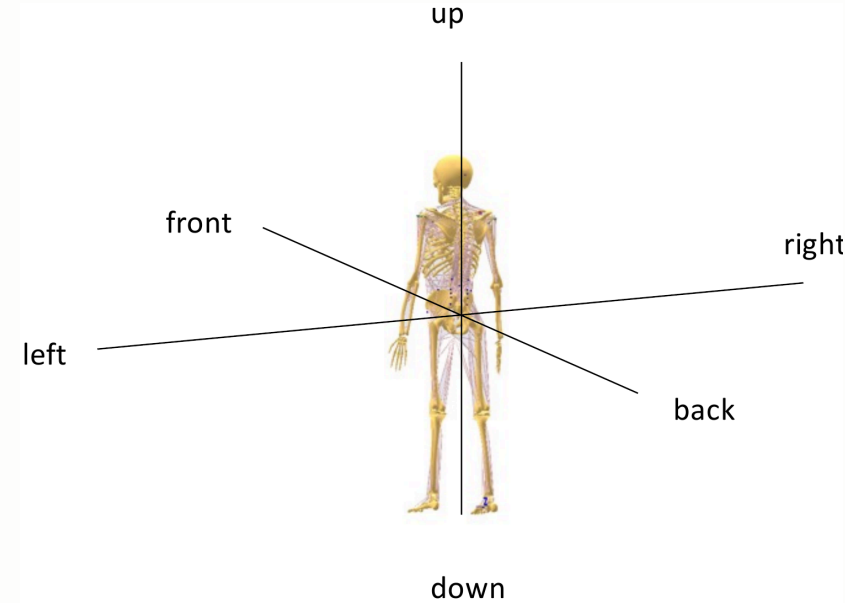
Moment of inertia about different axes

An object can turn about many axes, so it has a moment of inertia **for each one** — because the mass is spread out differently with respect to each.

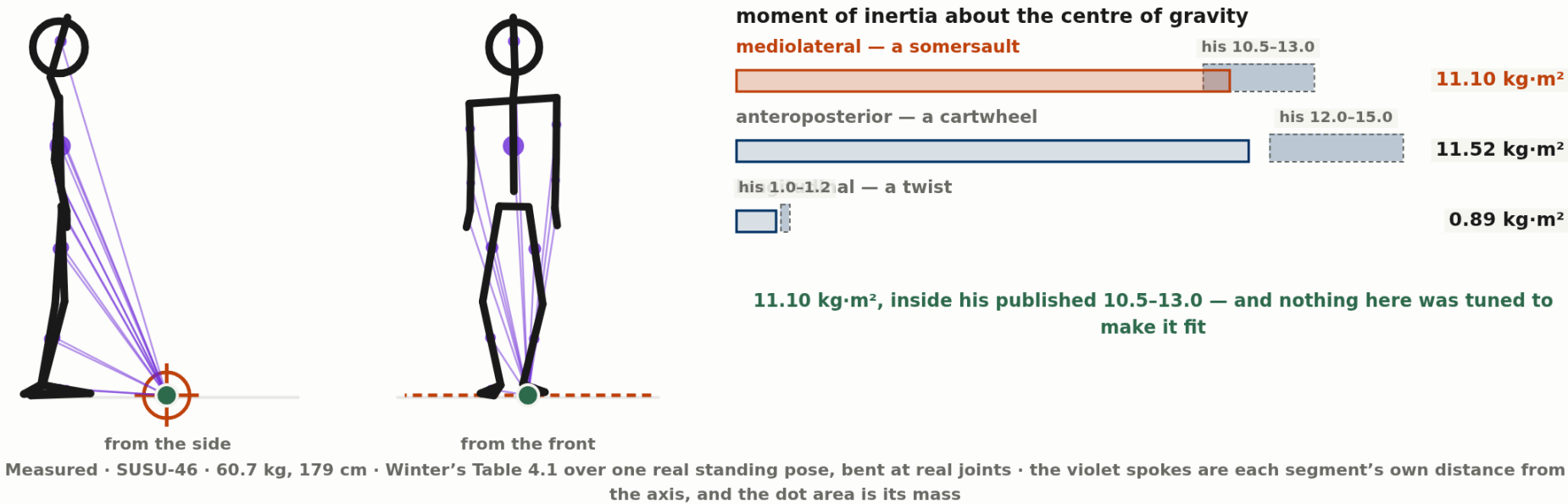
For a standing person the three that matter are:

- **mediolateral** — through the hips, side to side. A somersault turns about this one.
- **anteroposterior** — front to back. A cartwheel.
- **longitudinal** — head to toe. A twist.

They are not close to each other. For a standing adult the twist axis is roughly **ten times easier** to turn about than the other two — which is why a twist is the cheapest rotation in the air and a somersault the most expensive.



His slide 21, computed on a real body



Arms out

Knees bent

0°

0°

Hips bent

Trunk bent

0°

0°

Standing

Arms out

Loose tuck

Tight tuck

Somersault

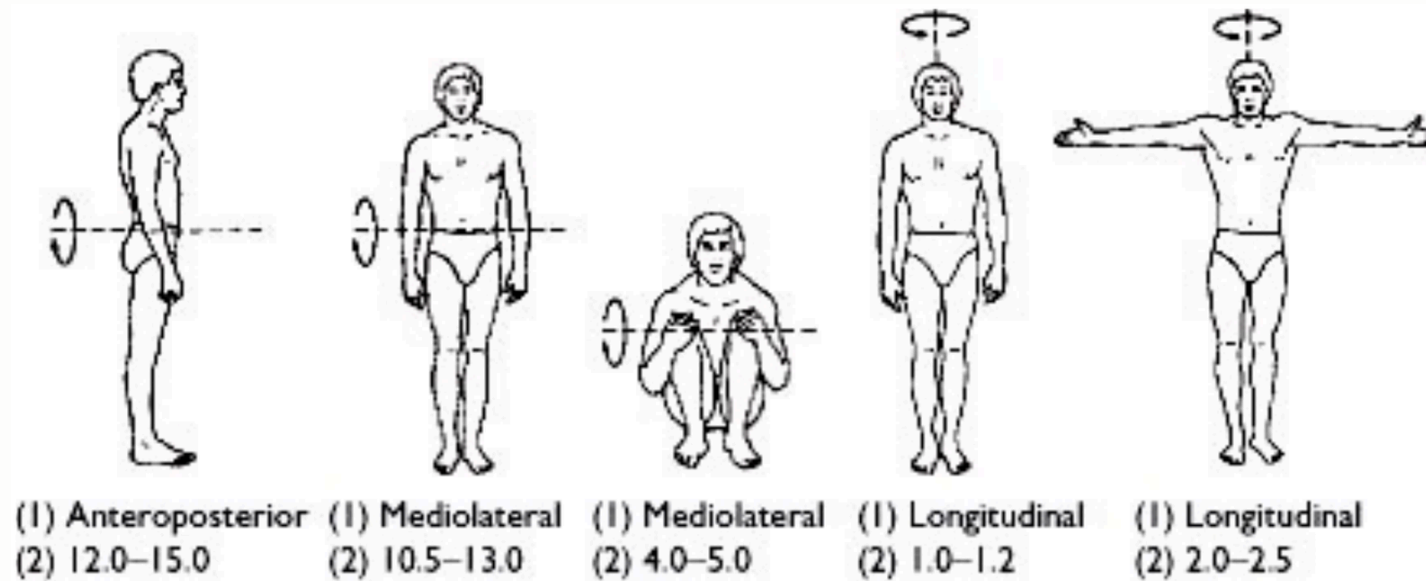
Cartwheel

Twist

His slide 21 gives five positions and a published moment of inertia for each. This is **one real measured body** bent into them and summed with Winter's table; the grey bands are his numbers. Put the arms out and the twist value doubles while the somersault value barely moves — the arms go out *along* one axis and *away from* the other.

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The published figure



Five positions, three axes, and the point of the whole lecture in one row of numbers: **the same person, from 1.0 to 15.0 kg·m²**, with nothing changed but posture and the choice of axis.

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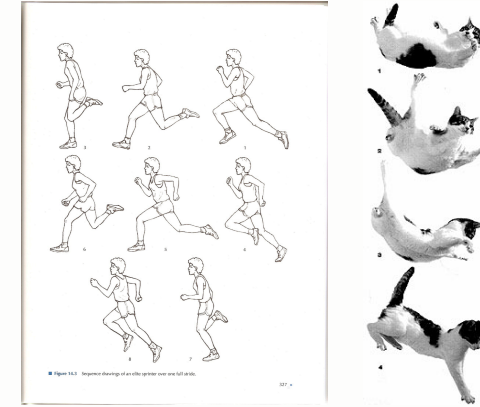
Manipulating the moment of inertia of the human body

Humans and animals change the orientation of their limbs constantly, and every change redistributes mass about every axis. The moment of inertia changes with it — not as a side effect, but as the thing being controlled.

Three examples, all on the next few slides:

- a sprinter folds the knee so the leg is cheaper to swing forward
- a diver tucks so the same angular momentum buys three somersaults instead of one
- a cat, with no angular momentum at all, still turns itself over

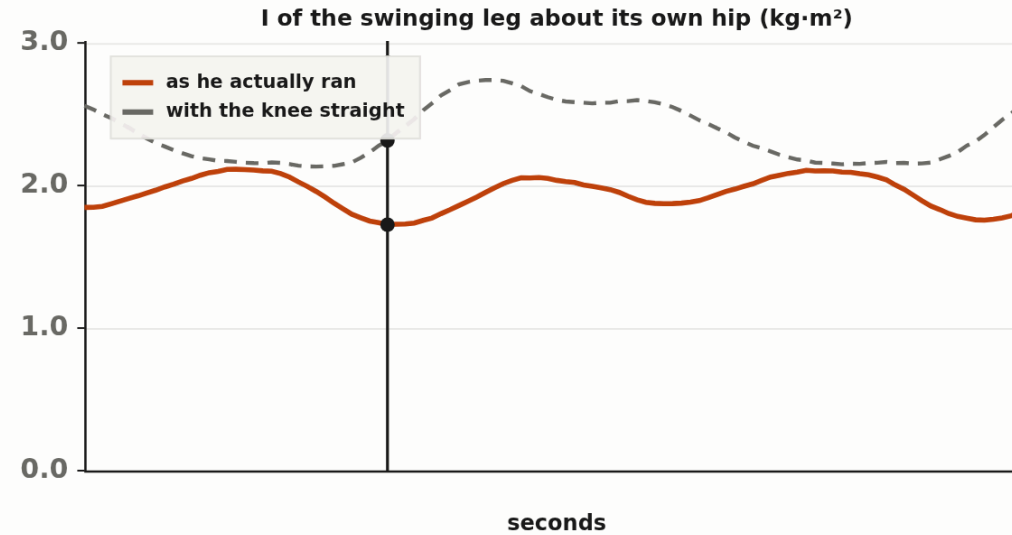
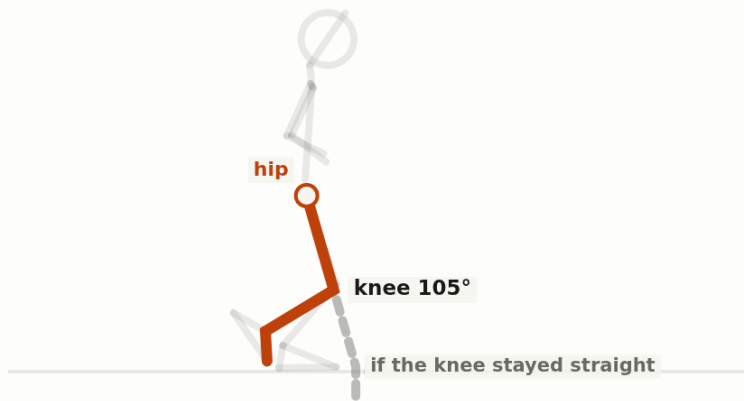
Only one quantity in $M = I\alpha$ is under voluntary control, and it is I . That is why so much technique is really about limb position.



The cat starts with zero angular momentum and lands feet-down with zero angular momentum. It never breaks the first law — it changes its moment of inertia about one axis while turning about another, twice, in opposite senses.

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A sprinter reduces the moment of inertia about the hip by flexing the knee



at this instant the bent leg is 25% easier to swing — 1.73 against 2.31 $\text{kg}\cdot\text{m}^2$

Measured · SUSU-46 running · 90 Hz · Winter's table on his own segment lengths · the knee works through 101-159° over this stride

Through the
stride

0.38 s

Show the straight-leg ghost

Hide it

▶ Run

His slide says a sprinter flexes the knee to reduce the moment of inertia about the hip. Here is a real runner doing it. The dashed ghost is the same leg with the knee straightened and the thigh left alone — the comparison the slide is making — and at the tightest point of swing the real leg is **26% easier to swing**. That is why the knee folds, and it costs nothing: the foot has nowhere to be until it lands.

Newton's first law, angular

Angular momentum, and the fact that nothing in the air can change it.

Angular momentum

In linear kinetics, an object in motion has **momentum** — inertia in motion:

$$p = m v$$

$$p = m (\text{inertia}) \cdot v (\text{in motion})$$

Angular momentum is the tendency of an object to remain in angular motion. It is determined by the moment of inertia and the angular velocity:

$$L = I \omega$$

$$L = I (\text{angular inertia}) \cdot \omega (\text{in angular motion})$$

Units: $\text{kg} \cdot \text{m}^2/\text{s}$.

The first law, angular. A body maintains a constant state of angular motion unless acted on by an external torque.

So in the air, with gravity the only force acting, and gravity acting *at* the centre of gravity:

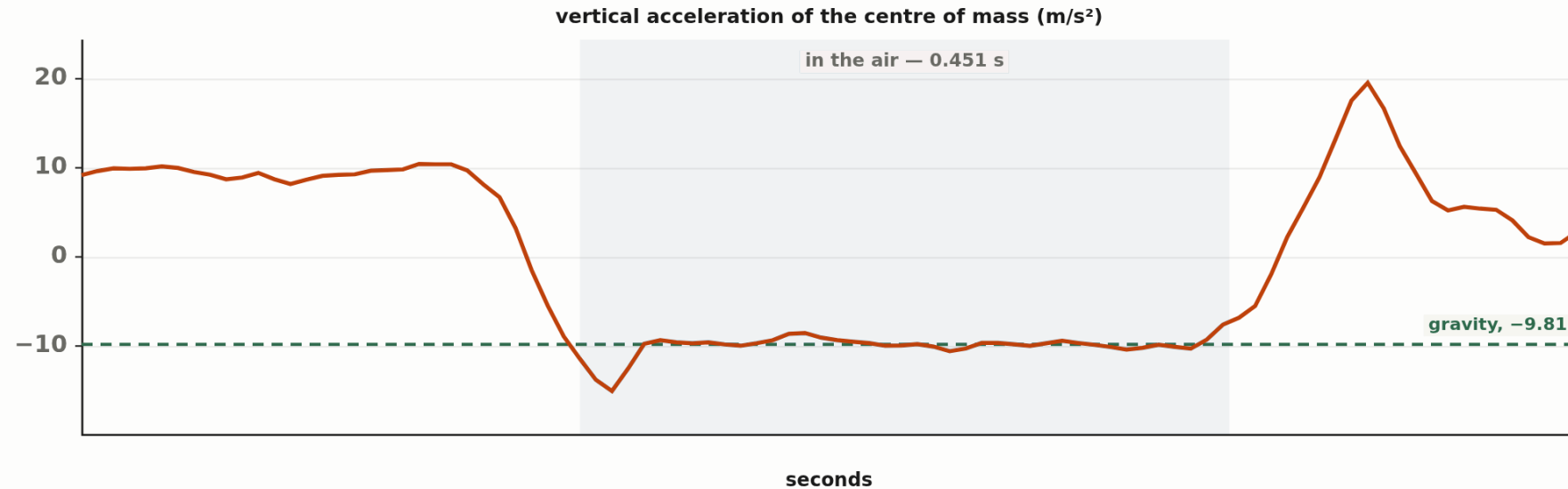
$$L_{\text{initial}} = L_{\text{final}}$$

$$I_i \omega_i = I_f \omega_f$$

— the **conservation of angular momentum**, and the reason a diver can decide in the air how many somersaults to do, but not whether to do any.

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Is that actually true? A measured flight phase



measured in flight: -9.71 ± 0.43 m/s², against gravity's -9.81 — one per cent, from markers alone. Nothing but gravity is acting.

Measured · SUSU-46 · 60.7 kg · a 14-segment Winter model over the markers, at 90 Hz · the shaded band is where the force plate reads zero, and the plate was not used to compute any of these three curves

What is acting on him

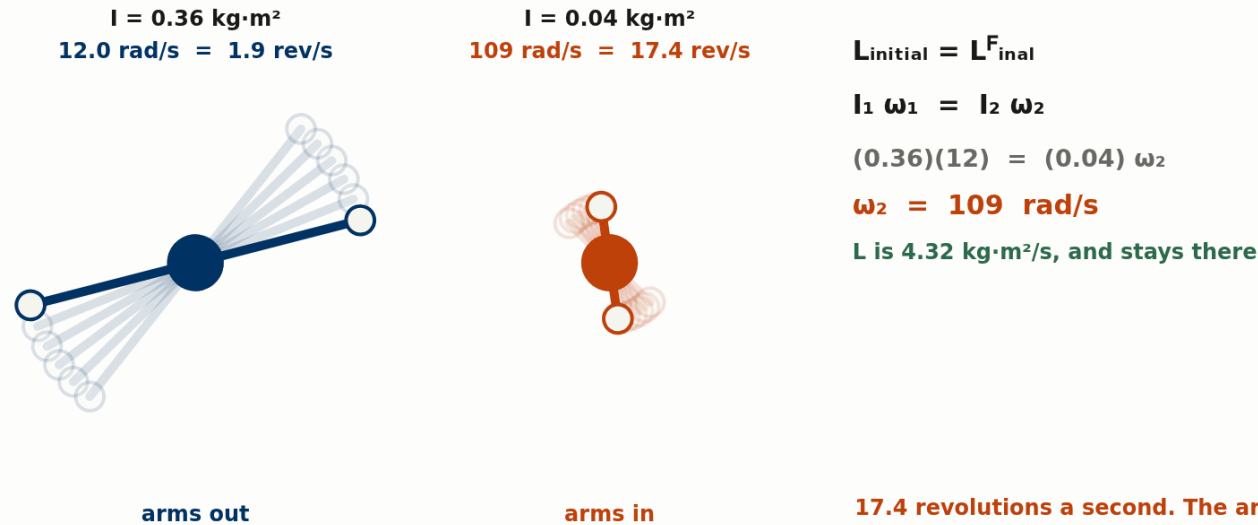
His angular momentum

How high he got

His slides say angular momentum is conserved in the air. The reason is that gravity pulls at the centre of mass, so it has no moment arm about the centre of mass, so there is no torque. That premise is measurable, and here it is measured: through the whole flight the centre of mass accelerates at **-9.71 m/s²**. Then look at his angular momentum — he has almost none, and nothing in the air can give him any.

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Conservation of angular momentum



$$L_{\text{initial}} = L_{\text{final}}$$

$$I_1 \omega_1 = I_2 \omega_2$$

$$(0.36)(12) = (0.04) \omega_2$$

$$\omega_2 = 109 \text{ rad/s}$$

L is $4.32 \text{ kg}\cdot\text{m}^2/\text{s}$, and stays there

17.4 revolutions a second. The arithmetic is right; the inputs are not. A 9.1 : 1 change in moment of inertia is more than a person can make — real skaters manage about 3.5 : 1, and the fastest spins ever filmed are near 5 rev/s.

Nothing is touching her, so nothing can change L · whatever I loses, ω makes up exactly · the two figures are drawn with their arms at the radius of gyration each moment of inertia implies

How far the
arms come in



His slide 27

A real skater

This lecture's measured body

Pull the arms in and the moment of inertia falls; angular momentum cannot, because nothing is touching her. So the angular velocity has to rise by exactly the factor the moment of inertia fell by. His own numbers are the first preset — the algebra is right and the answer is absurd, which is worth more than a tidy example. Compare the second.

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Example problem — the figure skater

If a figure skater has an angular velocity of **12 rad/s** with her arms extended ($I = 0.36 \text{ kg}\cdot\text{m}^2$), what does her angular velocity change to when she brings her arms in ($I = 0.04 \text{ kg}\cdot\text{m}^2$)?

Write down the proper equation

$$L_{\text{initial}} = L_{\text{final}}$$

$$I_i \omega_i = I_f \omega_f$$

Write down given and unknown values

$$I_i = 0.36 \text{ kg}\cdot\text{m}^2 \quad \omega_i = 12 \text{ rad/s}$$

$$I_f = 0.04 \text{ kg}\cdot\text{m}^2 \quad \omega_f = ?$$

Substitute and solve

$$(0.36)(12) = (0.04)\omega_f$$

$$\omega_f = \mathbf{108 \text{ rad/s}}$$

Check the answer against the world. 108 rad/s is **17.2 revolutions a second**. The fastest scratch spins on film are about **5**.

The algebra is right. The inputs are not: $0.36 \rightarrow 0.04$ is a **9 : 1** change in moment of inertia, and a person cannot do that. Measured skaters manage roughly **3.5 : 1** — which, from a realistic 9 rad/s start, lands near the 5 rev/s you can see.

Converting an answer into something you can picture is not an optional last step. It is how you find out whether you believe it.

rad/s $\rightarrow$ **rev/s**: divide by $2\pi \approx 6.28$. Worth doing every single time an angular velocity comes out of a problem.

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Example problem — the diver, with the radius of gyration

A **60 kg** diver leaves the board with a radius of gyration of **0.5 m** and an angular velocity of **4 rad/s**. What is his angular velocity in a tuck, with a radius of gyration of **0.25 m**?

Equations

$$I_i \omega_i = I_f \omega_f \quad I = m k^2$$

Moments of inertia in the two positions

$$I_1 = m k_1^2 = (60)(0.5)^2 = \mathbf{15 \text{ kg}\cdot\text{m}^2}$$

$$I_2 = m k_2^2 = (60)(0.25)^2 = \mathbf{3.75 \text{ kg}\cdot\text{m}^2}$$

Then conservation

$$(15)(4) = (3.75)\omega_f$$

$$\omega_f = \mathbf{16 \text{ rad/s}}$$

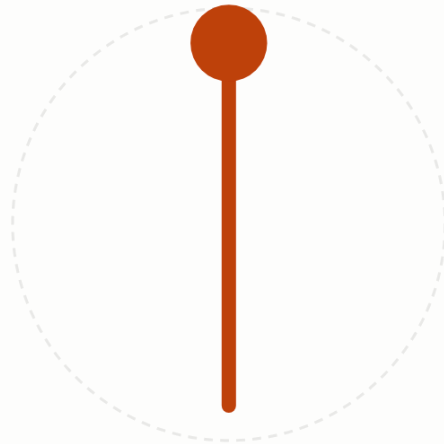


Halve k and you quarter I , because k is squared — so the angular velocity goes up by four, not two.

16 rad/s is 2.5 rev/s, which is what a tucked diver actually does. This one survives the sanity check.

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How many somersaults fit in the time you have



turned 0.00 of 1.00 revolutions

$$L = I \omega \rightarrow \omega = L / I$$

$$\omega = 39 / 7.450 = 5.23 \text{ rad/s}$$

$$\theta = \omega t = 5.23 \times 1.20 = 6.28 \text{ rad}$$

1.00 revolutions

a half

a single

a one and a half

a double

a two and a half

a triple

a three and a half

enough for a single

The Tutor's diver · $L = 39 \text{ kg}\cdot\text{m}^2/\text{s}$ and 1.2 s of flight, both fixed at take-off · layout $I = 7.45$ gives 1.00 rev, tuck $I = 2.475$ gives 3.01 · the only thing she still controls once her feet leave the board is how tightly she holds

How tightly
she tucks

$I = 7.45$

Time in the air

1.20 s

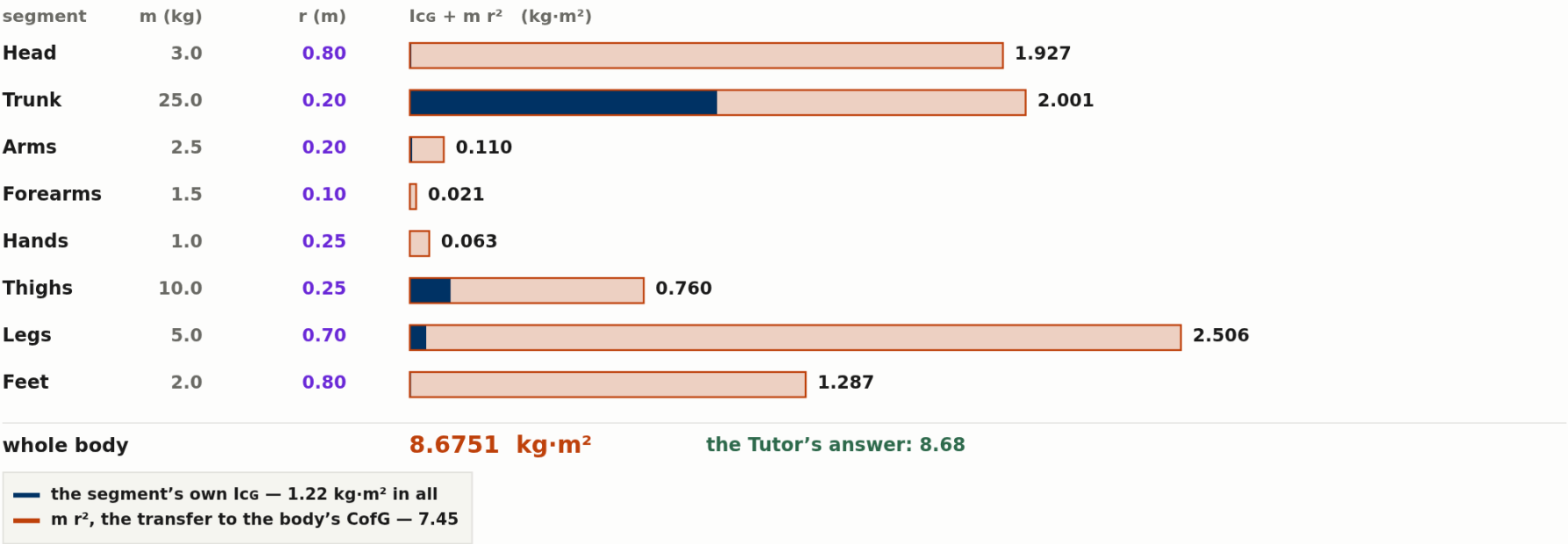
Layout

Tuck

▶ Dive

She leaves the board with a fixed angular momentum and a fixed time in the air — both decided before her feet leave. Everything after that is one choice: **how tightly to hold**. In the layout she completes a single somersault; in a tuck, with exactly the same push, she completes three.

The whole-body sum, segment by segment



Layout 8.68 against tuck 3.70 — a factor of 2.35, and all of it comes from four segments moving closer in

The Biomechanics Tutor, Moment of Inertia Q11 and Q12 · a 50 kg gymnast · the eight masses sum to exactly 50 kg · tucking changes r, not m and not l_{CG}

Layout

Tuck

Eight segments, each contributing its own moment of inertia *plus* its mass times its distance from the body's centre of gravity, squared. Switch to the tuck and watch which rows move: the head, the hands, the legs and the feet. The trunk does not move at all, because it is already at the centre — and it is half the body's mass.

Newton's second law, angular

What it takes to change any of this — and where in a movement you have to spend it.

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Newton's second law: the law of acceleration

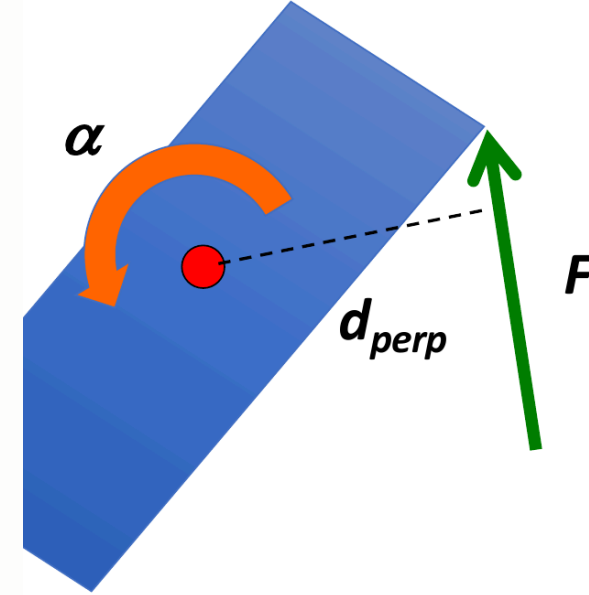
Linear. A force applied to a body causes an acceleration of that body, of a magnitude proportional to the force, in the direction of the force, and inversely proportional to the body's mass.

$$F = m a$$

Angular. A torque applied to a body causes an angular acceleration of that body, of a magnitude proportional to the torque, in the direction of the torque, and inversely proportional to the body's *moment of inertia*.

$$\tau = F d_{\perp} \quad M = I \alpha$$

The greater the inertia of an object, the more force — or torque — is needed to make it accelerate.

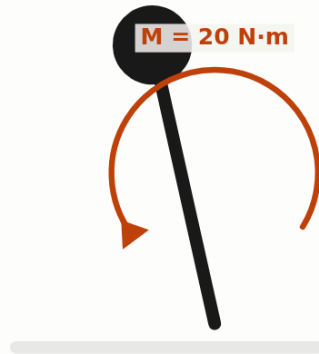


If a net external torque acts at a distance from the axis of rotation, the object accelerates angularly **in the direction of that torque**, in proportion to it, and in inverse proportion to its moment of inertia.

Two ways to get more angular acceleration out of the same muscle: push **harder**, or push **further from the**

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Angular impulse: a torque applied for a certain amount of time



angular impulse = $M \times t$

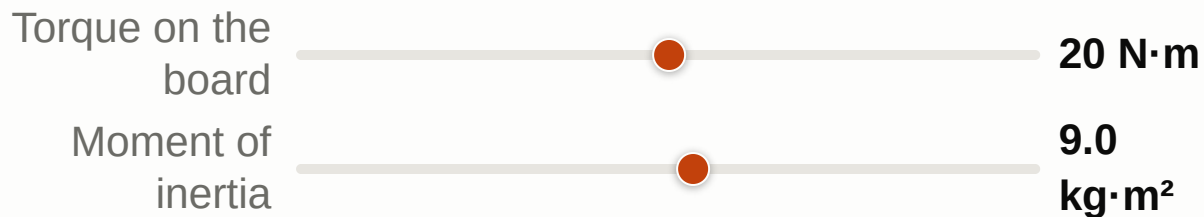


$$M t = \Delta L = I \omega$$

$$(20)(1.40) = (9.0) \omega$$

$$\omega = 3.11 \text{ rad/s} = 0.50 \text{ rev/s}$$

Everything the diver will ever have in the air is bought here, against the board · 20 N·m held for 1.40 s is 28.0 N·m·s of angular impulse, and that IS the angular momentum she leaves with · once her feet are off, no torque acts and the number is frozen



His numbers

► Push off

A torque held for a time is an **angular impulse**, and an angular impulse is a change in angular momentum — the same sentence as force, time and momentum, one lecture earlier. Note how little the diver gets: 20 N·m for a long 1.4 s buys 3.11 rad/s, half a turn a second. Everything else she does in the air

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The angular impulse–momentum relationship

A change in the angular momentum of a system depends on:

1. the **magnitude** of the external torques, and
2. the **time** over which those torques act.

Therefore an angular impulse causes a change in angular momentum:

$$M t = (I\omega)_f - (I\omega)_i$$

$$M t = \Delta L$$

SI units: **N·m·s**. Angular impulse is a vector — it has a sense, and torques in opposite senses subtract.

$\vec{J} = \vec{M}t$ $\vec{J} = (I\alpha)t$ $\vec{J} = I(\alpha t)$ $\vec{J} = I\omega$ $\vec{J} = \vec{L}$	$\vec{M} = I\alpha$
	$\vec{\alpha} = \frac{\vec{\omega}}{t}$ $\vec{\omega} = \vec{\alpha}t$



Each line is one substitution. Read it downward and it is a derivation, not a list to memorise.

Angular Kinetics 3

Example problem — the Olympic diver

An Olympic diver exerts a torque of **20 N·m** with his body in full extension ($I = 9 \text{ kg}\cdot\text{m}^2$) for **1.4 s** before leaving the platform. What is his angular velocity at the moment he leaves?

Given

$$M = 20 \text{ N}\cdot\text{m} \quad t = 1.4 \text{ s} \quad I = 9 \text{ kg}\cdot\text{m}^2$$

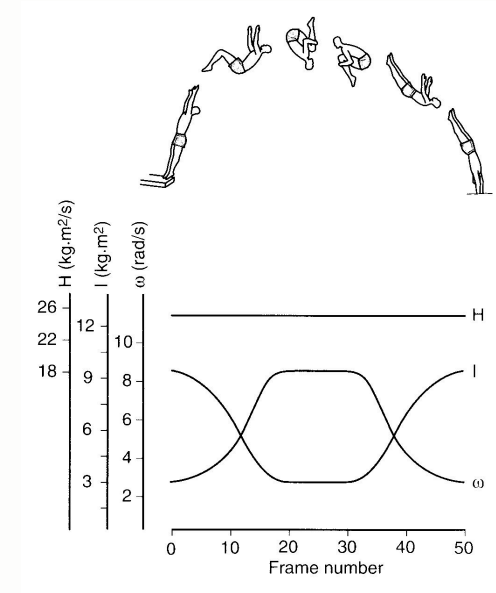
Solution

$$M t = (I\omega)_f - (I\omega)_i$$

$$(20)(1.4) = (9)\omega - (9)(0)$$

$$28 = 9\omega$$

$\omega = \mathbf{3.11 \text{ rad/s}}$, in the direction the torque was applied



The published figure, and the whole lecture in one picture: through the dive **L is a flat line** while **I** and **ω** cross over each other.

What to take away

- $I = \Sigma m r^2$ — mass counts once, distance counts twice. Distribution beats mass.
- $I = m k^2$ — the radius of gyration is the distance at which all the mass would give the same I , and it is how segment tables are written.
- $I_{\text{axis}} = I_{\text{CofG}} + m d^2$ — which is why the centre of gravity is always the cheapest axis, and why a free body turns about it.
- **One I per axis per posture.** For one person, 1.0 to 15.0 kg·m².
- $L = I\omega$, and in the air L is frozen — gravity acts at the centre of gravity, so it has no moment arm about it. Measured: -9.71 m/s^2 and no torque.
- $M = I\alpha$, and $Mt = \Delta L$ — angular momentum is bought on the ground and spent in the air.
- **Check your answers against the world.** 108 rad/s is 17 rev/s, and nobody spins that fast.

The through-line: of everything in $M = I\alpha$ and $L = I\omega$, the only term a person can change mid-movement is I . That is why technique looks the way it does.